



Mathematics Part 1 — Cbse Periodic Test (Pt) — 40 Marks

ANSWER KEY — FOR GRADING

Subject: Mathematics Part 1 | Class: Class 9

Total Marks: 40 | Time Allowed: 1h 30m | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 8 questions. Each carries 1 mark(s).		
1	Which identity can be used to expand the product of two binomials of the form $(ax + b)(cx + d)$? (A) $(ax + b)(cx + d) = acx^2 + (ad + bc)x + bd$ (B) $(ax + b)(cx + d) = acx^2 + (ab + cd)x + bd$ (C) $(ax + b)(cx + d) = acx^2 + (ad - bc)x + bd$ (D) $(ax + b)(cx + d) = acx^2 + (ad + bc)x - bd$ Answer: Correct Option: (A) ► Why? A. The standard identity for the product of two general binomials is $(ax + b)(cx + d) = acx^2 + (ad + bc)x + bd$. [Ch 4: Exploring Algebraic Identities]	1
2	The centre of the unique circle passing through three non-collinear points is called the: (A) Incentre (B) Circumcentre (C) Orthocentre (D) Centroid Answer: Correct Option: (B) ► Why? B. The centre of the circle passing through three non-collinear points is called the circumcentre of the three points. [Ch 5: I'm Up and Down, and Round and Round]	1



Q.NO.	QUESTIONS	MARKS
3	One of the columns on the Ishango bone groups notches into which of the following sets of numbers? (A) 2, 4, 8, 16 (B) 11, 13, 17, 19 (C) 1, 3, 5, 7 (D) 10, 12, 14, 16 Answer: Correct Option: (B) ► Why? B. One of the columns on the Ishango bone groups notches into 11, 13, 17, and 19, which are the prime numbers between 10 and 20. [Ch 3: The World of Numbers]	1
4	The algebraic expression $36x^2 + 12x + 1$ is written in the form $a^2 + 2ab + b^2$ to factorise it using the identity $(a + b)^2 = a^2 + 2ab + b^2$. What are the values of a and b in this case? (A) $a = 36x$, $b = 1$ (B) $a = 6x$, $b = 1$ (C) $a = 6$, $b = 1$ (D) $a = 6x$, $b = 12$ Answer: Correct Option: (B) ► Why? B. The expression $36x^2 + 12x + 1$ can be written as $(6x)^2 + 2(6x)(1) + 1^2$, which matches $a^2 + 2ab + b^2$ with $a = 6x$ and $b = 1$. [Ch 4: Exploring Algebraic Identities]	1
5	Which term of the sequence 2, $2\sqrt{2}$, 4, ... is 128? (A) 14th (B) 15th (C) 16th (D) 13th Answer: Correct Option: (D) ► Why? D. The sequence is a GP with $a=2$ and $r=\sqrt{2}$. Using $t_n = 2(\sqrt{2})^{n-1} = 128$, we get $(\sqrt{2})^{n-1} = 64 = 2^6 = (\sqrt{2})^{12}$, so $n-1=12$ and $n=13$. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	1
6	What is the side of a cube whose volume is $p^3 + 6p^2q + 12pq^2 + 8q^3$ cubic units? (A) $p + q$ (B) $p + 2q$ (C) $2p + q$ (D) $p + 8q$ Answer: Correct Option: (B)	1



Q.NO.	QUESTIONS	MARKS
	► Why? B. The volume can be rewritten as $(p)^3 + 3(p)^2(2q) + 3(p)(2q)^2 + (2q)^3 = (p + 2q)^3$, so the side is $p + 2q$. [Ch 4: Exploring Algebraic Identities]	
7	Assertion (A): In an arithmetic progression, each term after the first is obtained by adding a fixed number to the previous term. Reason (R): The fixed number added in an arithmetic progression is called the common difference. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (A) ► Why? A. The assertion is true because an arithmetic progression is defined as a list of numbers where each term after the first is obtained by adding a fixed number to the previous term. The reason correctly explains the assertion by identifying this fixed number as the common difference, which is the defining feature of an AP. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	1
8	Assertion (A): The expression $10x - x^2$ is a linear function. Reason (R): The expression $2x + 3$ is a linear function. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. Answer: Correct Option: (D) ► Why? D. A is false but R is true. The expression $10x - x^2$ is a quadratic function, not a linear one, whereas $2x + 3$ is explicitly identified as a linear function. [Ch 2: Introduction to Linear Polynomials]	1
Section B Attempt all 4 questions. Each carries 2 mark(s).		
9	A pattern of square tiles starts with 1 tile in Stage 1, 3 tiles in Stage 2, and 5 tiles in Stage 3. Write the linear expression for the number of tiles at Stage n. Answer: The number of tiles at Stage n is given by $2n - 1$. This is because the number of squares at each stage is one less than twice the stage number, and the difference between consecutive terms is 2. ► Why? The expression is derived by observing that the number of tiles is one less than twice the stage number, forming a linear pattern. [Ch 2: Introduction to Linear Polynomials]	2
10	How many circles can be drawn passing through three non-collinear points?	2



Q.NO.	QUESTIONS	MARKS
	Answer: Exactly one unique circle can be drawn passing through three non-collinear points. The centre of this circle is the circumcentre, which lies at the intersection of the perpendicular bisectors of the line segments joining the points. ► Why? The intersection of the perpendicular bisectors of two line segments connecting the three points yields a single point equidistant from all three, forming the unique centre. [Ch 5: I'm Up and Down, and Round and Round]	
11	A chord of length 24 cm is drawn in a circle of diameter 26 cm. Find the distance from the centre of the circle to the chord. Answer: The radius is 13 cm. The perpendicular from the centre bisects the chord, forming a right triangle with half the chord (12 cm) and the radius (13 cm). The distance from the centre is $\sqrt{(13^2 - 12^2)} = \sqrt{(169 - 144)} = 5$ cm. ► Why? The perpendicular from the centre to the chord bisects it, creating a right triangle where the radius is the hypotenuse, allowing the use of the Pythagorean theorem. [Ch 5: I'm Up and Down, and Round and Round]	2
12	Are rational numbers closed under multiplication? Justify your answer. Answer: Yes, rational numbers are closed under multiplication. When two rational numbers are multiplied, the result is again a rational number. Using the rule $a/b \times c/d = ac/bd$, since ac and bd are integers and $bd \neq 0$, the product is rational. ► Why? The source states that rational numbers are closed under multiplication and provides the multiplication rule that confirms the product remains a rational number. [Ch 3: The World of Numbers]	2
Section C Attempt 3 out of 4.		
13	The first term of an AP is 11 and the common difference is -3. Find the 10th term of this AP.	3



Q.NO.	QUESTIONS	MARKS
	Answer: Given: An Arithmetic Progression (AP) with first term $a = 11$ and common difference $d = -3$. Formula: The n^{th} term of an AP is given by $t_n = a + (n - 1)d$. Substitution: To find the 10th term, substitute $n = 10$: $t_{10} = 11 + (10 - 1)(-3)$ Calculation: $t_{10} = 11 + 9 \times (-3)$ $t_{10} = 11 - 27$ $t_{10} = -16$ $\therefore$ The 10th term of the AP is -16. The negative value indicates the sequence is decreasing, which is consistent with a negative common difference. ► Why? The 10th term of an AP is found using the formula $t_n = a + (n-1)d$ with $a=11$ and $d=-3$, yielding -16. ► Step-by-step: - Step 1: Identify the given values: first term $a = 11$, common difference $d = -3$, and $n = 10$. - Step 2: Apply the formula for the nth term of an AP: $t_n = a + (n - 1)d$. - Step 3: Substitute the values and calculate: $t_{10} = 11 + (10 - 1)(-3) = 11 - 27 = -16$. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	
	OR In a circle, a chord is 5 cm away from the centre. If the radius of the circle is 13 cm, what is the length of the chord? Answer: Given: Distance from centre to chord = 5 cm, Radius = 13 cm. Formula: A perpendicular from the centre to a chord bisects it. By the Pythagorean Theorem, $r^2 = d^2 + (c/2)^2$. Substitution: $13^2 = 5^2 + (c/2)^2$ $169 = 25 + (c/2)^2$ $144 = (c/2)^2$ $c/2 = 12 \text{ cm}$ Answer: $c = 24 \text{ cm}$. $\therefore$ The length of the chord is 24 cm.	



Q.NO.	QUESTIONS	MARKS
	► Why? The perpendicular from the centre bisects the chord, forming a right triangle where the radius is the hypotenuse, allowing the use of the Pythagorean theorem. ► Step-by-step: - Step 1: Note that the perpendicular from the centre to the chord bisects it, forming a right-angled triangle with the radius, half-chord, and distance from centre. - Step 2: Apply the Pythagorean theorem: $r^2 = d^2 + (\text{half-chord})^2$. - Step 3: Substitute the given values: $13^2 = 5^2 + (\text{half-chord})^2$, so $169 = 25 + (\text{half-chord})^2$. - Step 4: Solve for the half-chord: $(\text{half-chord})^2 = 144$, so half-chord = 12 cm. The full chord length is 24 cm. [Ch 5: I'm Up and Down, and Round and Round]	
14	For a linear equation of the form $y = ax$ where $a > 0$, how does the steepness of the straight line change when $a > 1$ compared to when $a < 1$? Where do all such lines pass through? Answer: Given: Linear equations of the form $y = ax$ where $a > 0$. Steepness: When $a > 1$, the line is steeper than the line $y = x$ (which is equally inclined to both axes). When $a < 1$, the line is less steep than the line $y = x$. The value of a represents the slope of the line. Common Point: All straight lines representing an equation of the form $y = ax$ always pass through the origin, which is the point $(0, 0)$. This is because substituting $x = 0$ gives $y = 0$. ► Why? The coefficient 'a' in $y = ax$ acts as the slope. A slope greater than 1 makes the line rise faster (steeper), while a slope less than 1 makes it rise slower (less steep). Since the y-intercept is 0, all these lines intersect at the origin. ► Step-by-step: - Step 1: Identify that in $y = ax$, 'a' represents the slope of the line. - Step 2: Recall that when $a > 1$, the line is steeper than $y = x$, and when $a < 1$, it is less steep. - Step 3: State that all lines of this form pass through the origin $(0, 0)$. [Ch 2: Introduction to Linear Polynomials]	3
15	Write the identity for the product of $(x + y)$ and $(x^2 - xy + y^2)$. Answer: Given: Expression $(x + y)(x^2 - xy + y^2)$ Identity: The product of $(x + y)$ and $(x^2 - xy + y^2)$ results in the sum of cubes. Calculation: $(x + y)(x^2 - xy + y^2) = x^3 + y^3$ ∴ The identity is $(x + y)(x^2 - xy + y^2) = x^3 + y^3$. This identity is extremely useful for quickly factoring the sum of two perfect cubes without performing long multiplication.	3



Q.NO.	QUESTIONS	MARKS
	► Why? Multiplying a binomial by a specific trinomial yields the sum of cubes, which is a standard algebraic identity used for factorisation. ► Step-by-step: - Step 1: Recognize the pattern for the sum of cubes involving the product of $(x + y)$ and $(x^2 - xy + y^2)$. - Step 2: State the product directly as $x^3 + y^3$. - Step 3: Write the complete identity: $(x + y)(x^2 - xy + y^2) = x^3 + y^3$. [Ch 4: Exploring Algebraic Identities]	
Section D Attempt 2 out of 3.		
16	Find the sum of the consecutive integers from 25 to 58 by using the formula for the sum of the first n natural numbers. Show the step-by-step calculation.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Given: The sequence of consecutive integers from 25 to 58. Formula: The sum of the first n natural numbers is given by: $S_n = (n(n + 1))/2$ Explanation: The sum of the first n natural numbers can be found using the formula derived from visual representations. This formula is highly useful for finding the sum of a series of consecutive integers without adding them one by one. To find the sum of a specific range of consecutive integers, such as from 25 to 58, we can subtract the sum up to the lower limit from the sum up to the upper limit. This method relies on the fact that: $25 + 26 + \dots + 58 = (1 + 2 + \dots + 58) - (1 + 2 + \dots + 24)$ By applying the formula $S_n = (n(n+1))/2$, we can easily compute both these sums and find their difference. Substitution: $S_{58} = (58 \times 59)/2$ $S_{24} = (24 \times 25)/2$ Calculation: $S_{58} = 29 \times 59 = 1711$ $S_{24} = 12 \times 25 = 300$ Sum from 25 to 58 = $S_{58} - S_{24}$ $\text{Sum} = 1711 - 300 = 1411$ ∴ The sum of the consecutive integers from 25 to 58 is 1411. ► Why? The sum of a range of consecutive integers is found by subtracting the sum of natural numbers up to the lower limit from the sum up to the upper limit, using the formula: $S_n = (n(n+1))/2$ ► Step-by-step: 1. Step 1: Express the sum from 25 to 58 as the difference of two sums of natural numbers ($S_{58} - S_{24}$). 2. Step 2: Apply the formula $S_n = (n(n+1))/2$ to find S_{58} and S_{24}. 3. Step 3: Calculate the values of S_{58} and S_{24} by simplifying the fractions. 4. Step 4: Subtract S_{24} from S_{58} to get the final sum. [Ch 8: Predicting What Comes Next: Exploring Sequences and Progressions]	



Q.NO.	QUESTIONS	MARKS
	OR State the identity for $(a + b + c)^2$ and describe its geometric interpretation using a square of side $(a + b + c)$. Use this identity to find the value of 119^2 and to expand $(3x - 2y + 4z)^2$.	



Q.NO.	QUESTIONS	MARKS
	Answer: The identity for the square of the sum of three numbers is: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Geometric Interpretation: A square of side $(a + b + c)$ is drawn. The side is divided into three segments of lengths a, b, and c. Lines parallel to the sides are drawn at these division points, creating a grid of 9 regions inside the square. These regions consist of: • Three squares with areas a^2, b^2, and c^2. • Six rectangles — two each with areas ab, bc, and ca. The total area of the big square equals the sum of all these regions, giving the identity: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Finding 119^2: Given: $119 = 100 + 10 + 9$, so $a = 100$, $b = 10$, $c = 9$. Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Substitution: $119^2 = (100 + 10 + 9)^2$ $= 100^2 + 10^2 + 9^2 + 2(100)(10) + 2(100)(9) + 2(10)(9)$ $= 10000 + 100 + 81 + 2000 + 1800 + 180$ $= 14161$ $\therefore 119^2 = 14161$ Expanding $(3x - 2y + 4z)^2$: Given: $a = 3x$, $b = -2y$, $c = 4z$ Formula: $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ Substitution: $(3x - 2y + 4z)^2 = (3x)^2 + (-2y)^2 + (4z)^2 + 2(3x)(-2y) + 2(-2y)(4z) + 2(4z)(3x)$ $= 9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24xz$	



Q.NO.	QUESTIONS	MARKS
	$\therefore (3x - 2y + 4z)^2 = 9x^2 + 4y^2 + 16z^2 - 12xy - 16yz + 24xz$ ► Why? The identity $(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ arises from partitioning a square of side $(a+b+c)$ into 3 smaller squares and 6 rectangles, and it simplifies both numerical squaring and algebraic expansion. ► Step-by-step: Step 1: State the identity $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$. Step 2: Describe the geometric model — a square of side $(a+b+c)$ divided into 3 squares and 6 rectangles, and label each region with its area. Step 3: Apply the identity to find 119^2 by expressing 119 as $100 + 10 + 9$ and substituting into the formula. Step 4: Expand $(3x - 2y + 4z)^2$ by identifying $a = 3x$, $b = -2y$, $c = 4z$ and carefully substituting while tracking signs. Step 5: Simplify all terms and state the final expanded expression. [Ch 4: Exploring Algebraic Identities]	
17	The diameter of a circle is 26 cm. A chord of length 24 cm is drawn in the circle. Find the distance of the chord from the centre. Also, a cyclic quadrilateral ABCD is inscribed in this circle with $\angle A = 75^\circ$ and $\angle B = 110^\circ$. Find the measures of $\angle C$ and $\angle D$.	5



Q.NO.	QUESTIONS	MARKS
	Answer: Part (a): Distance of chord from the centre Given: Diameter = 26 cm, Chord length = 24 cm Radius: $r = 26/2 = 13 \text{ cm}$ Property used: A perpendicular from the centre to a chord bisects the chord. Half of chord: $24/2 = 12 \text{ cm}$ Right triangle formed by radius, half-chord, and distance from centre: $d^2 + 12^2 = 13^2$ $d^2 + 144 = 169$ $d^2 = 169 - 144 = 25$ $d = \sqrt{25} = 5 \text{ cm}$ $\therefore$ Distance from centre to chord = 5 cm Part (b): Angles of cyclic quadrilateral ABCD Given: $\angle A = 75^\circ$, $\angle B = 110^\circ$, ABCD is a cyclic quadrilateral Property used: In a cyclic quadrilateral, opposite angles sum to 180°. Finding $\angle C$: $\angle A + \angle C = 180^\circ$ $75^\circ + \angle C = 180^\circ$ $\angle C = 180^\circ - 75^\circ = 105^\circ$ $\therefore \angle C = 105^\circ$ Finding $\angle D$: $\angle B + \angle D = 180^\circ$ $110^\circ + \angle D = 180^\circ$	



Q.NO.	QUESTIONS	MARKS
	$\angle D = 180^\circ - 110^\circ = 70^\circ$ $\therefore \angle D = 70^\circ$ Verification: Sum of all four angles = $75^\circ + 110^\circ + 105^\circ + 70^\circ = 360^\circ$ ✓ Diagram description: Draw a circle with centre O. Draw chord of length 24 cm and show the perpendicular from O bisecting it. Mark the radius (13 cm), half-chord (12 cm), and distance (5 cm). Separately, draw cyclic quadrilateral ABCD inscribed in a circle, labelling all four angles. ► Why? The perpendicular from the centre bisects the chord, creating a right triangle with the radius, half-chord, and distance from centre, solved using the Pythagorean theorem. Opposite angles of a cyclic quadrilateral are supplementary (sum to 180°). ► Step-by-step: Step 1: Find the radius from the diameter: $r = 26/2 = 13$ cm. Step 2: Use the property that perpendicular from centre bisects the chord; half-chord = $24/2 = 12$ cm. Step 3: Apply the Pythagorean theorem in the right triangle: $d^2 + 12^2 = 13^2$, solve for $d = 5$ cm. Step 4: For the cyclic quadrilateral, use the property that opposite angles sum to 180° to find $\angle C = 105^\circ$ and $\angle D = 70^\circ$. Step 5: Verify that all four angles sum to 360° and describe the diagram. [Ch 5: I'm Up and Down, and Round and Round]	
Section E Attempt 1 out of 1.		
18	Case Study: Book Club Subscriptions and Linear Polynomials A book club charges a joining fee of ₹100 and ₹20 for every book borrowed. The total amount a member pays is represented by the linear polynomial $P(b) = 20b + 100$, where b is the number of books. The amount increases by a constant value for each additional book borrowed, representing linear growth. Another club charges no joining fee but ₹40 per book, represented by the polynomial $Q(b) = 40b$. Both $P(b)$ and $Q(b)$ are examples of univariate polynomials. Polynomials can also be thought of as input-output processes, where substituting a value for the variable gives an output. (i) What is the degree of the polynomial $P(b) = 20b + 100$? (ii) Identify the constant term of the polynomial $P(b) = 20b + 100$. (iii) Polynomials can be thought of as input-output processes. Calculate the total amount a member of the first club will pay if they borrow 5 books. (iv) Does the first club's plan represent linear growth or linear decay? Justify your answer based on the slope of the line. Answer: (i) The degree of the polynomial $P(b) = 20b + 100$ is 1. (ii) The constant term of the polynomial $P(b) = 20b + 100$ is 100. (iii) Substitute $b = 5$ in $P(b) = 20b + 100$. $P(5) = 20(5) + 100 = 100 + 100 = 200$. The total amount is ₹200. (iv) It represents linear growth because it is represented by a straight line with a positive slope (the coefficient of b is 20, which is positive).	5



Q.NO.	QUESTIONS	MARKS
	► Why? i) The degree of the polynomial $P(b) = 20b + 100$ is 1. ii) The constant term of the polynomial $P(b) = 20b + 100$ is 100. iii) Substitute $b = 5$ in $P(b) = 20b + 100$. $P(5) = 20(5) + 100 = 100 + 100 = 200$. The total amount is ₹200. iv) It represents linear growth because it is represented by a straight line with a positive slope (the coefficient of b is 20, which is positive). [Ch 2: Introduction to Linear Polynomials]	

